2024AC05653 D. Mallikarjuna Reddy

ISM Assignment - 2

Q1 : Light bulbs — life not normally distributed. But it has mean of 1200 hrs. std dev = 100 hrs. Random sample of 36 bulbs taken, probability of life span more than 1220 hrs. ?

If independent observations don't follow normal distributions then the averages follows Normal, if n is large. [Central Limit Theorm.]

Given : $\mu = 1200$

$\sigma = 100$

$n = 36$

$P(\bar{X} > 1220) = ?$

$n > 30 \Rightarrow$ so C.L.T $\Rightarrow$ $\bar{X} \sim \text{Normal}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

Standard Error $\Rightarrow$ deviation of Error.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{100}{\sqrt{36}} = \frac{100}{6} = 16.66$$

$P(\bar{X} > 1220) = ?$

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{1220 - 1200}{16.6667} = \frac{20}{16.6667} = 1.20$$

$$P(\bar{X} > 1220) = P(Z > 1.20) = 1 - P(Z < 1.20)$$
$$= 1 - 0.8849$$
$$= 0.1151 \Rightarrow 11.5\%$$

$\therefore$ 11.5%, a sample will have average 1220 hrs above.

Q2 : —— 2024AC05653 —— D. Mallikarjuna Reddy ——

Avg. daily protien intake is at least 60 grams. A sample of 25, showed mean of 57 grams, with std. dev 8 grams. At the 5% significant level, test whether avg intake is at least 60 grams?

Given : $n = 25$, $\bar{x} = 57$, $S = 8$ } sample data

$\mu = 60$ } population claim:

$\alpha = 5\% = 0.05$

Claim: Average intake is atleast 60 gm. $\Rightarrow \geq 60$

So, $H_0 : \mu \geq 60$

alternative $H_a : \mu < 60$

We don't know population σ, so we chose t-test.

$$t\text{-statistic} = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

$$S.E = S/\sqrt{n} = 8/\sqrt{25} = 8/5 = 1.6$$

$$t\text{-statistic} = \frac{57 - 60}{1.6} = -1.875$$

degrees of freedom (df) $= n-1 = 24$

∴ From t-table for $\alpha = 0.05$ & $df = 24$ $\Rightarrow -1.711$

$t < t_{critical} \Rightarrow$ reject H_0

$-1.875 < -1.711 \Rightarrow$ TRUE $\Rightarrow$ reject H_0.

∴ From the given data (evidence), avg $\underbrace{\text{intake is less than 60 gms}}_{H_a}$.

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(Q3):

channel	satisfaction scores
Email	6.5, 7.0, 6.8, 6.9, 7.1
Social Media	7.8, 8.0, 7.5, 7.9, 8.1
Direct Mail	6.2, 6.4, 6.0, 6.1, 6.3

one-way ANOVA to test whether mean scores differ b/w channels at 5% significant level.

$$Email\ (\bar{x}_1) = 6.5+7.0+6.8+6.9+7.1/5 = 34.3/5 = 6.86$$

$$S.M.\ (\bar{x}_2) = 7.8+8.0+7.5+7.9+8.1/5 = 39.3/5 = 7.86$$

$$D.M\ (\bar{x}_3) = 6.2+6.4+6.0+6.1+6.3/5 = 31.0/5 = 6.20$$

$$\bar{\bar{x}} = 34.3+39.3+31.0/5+5+5 = 6.973$$

ANOVA ⇒ works by comparing the variance between the groups to the variance within the groups.

Test ⇒ $H_0: \mu_{Email} = \mu_{S.M} = \mu_{D.M}$

H_a: At least one of the mean different.

Sum of Squares between the groups (SS_B):

* Sum of Squares between the given groups 3. (it measures the variation between

$$SS_B = n_1(\bar{x}_1 - \bar{\bar{x}})^2 + n_2(\bar{x}_2 - \bar{\bar{x}})^2 + n_3(\bar{x}_3 - \bar{\bar{x}})^2$$

$$= 5(6.86 - 6.97)^2 + 5(7.86 - 6.97)^2 + 5(6.20 - 6.97)^2$$

$$= 5(0.0127) + 5(0.786) + 5(0.597)$$

$$= 0.0635 + 3.93 + 2.985$$

$$SS_B = 6.977 \quad \boxed{\sum_{i=1}^{k} n_i(\bar{x}_i - \bar{\bar{x}})^2}$$

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* Sum of Squares within Groups(SS_W):

(measures variation within each channel.

$$SS_W = \sum (x_{i1}-\bar{x}_1)^2 + \sum (x_{i2}-\bar{x}_1)^2 + \dots + \sum (x_{ik}-\bar{x}_1)^2$$

SSW (Email) $= (6.5-6.86)^2 + (7.0-6.86)^2 + (6.8-6.86)^2 + (6.9-6.86)^2 + (7.1-6.86)^2$

$= 0.1296 + 0.0196 + 0.0036 + 0.0036 + 0.0576 = 0.214$

SSW (S.M) $= (7.8-7.86)^2 + (8.0-7.86)^2 + (7.5-7.86)^2 + (7.9-7.86)^2 + (8.1-7.86)^2$

$= 0.0036 + 0.0196 + 0.0961 + 0.0081 + 0.0576 = 0.185$

SSW (D.M) $= (6.2-6.20)^2 + (6.4-6.20)^2 + (6.0-6.20)^2 + (6.1-6.20)^2 + (6.3-6.20)^2$

$= 0 + 0.04 + 0.04 + 0.01 + 0.01 = 0.10$

$SS_W = 0.214 + 0.185 + 0.10 = 0.499$

* Calculating Mean Squares $= \dfrac{\text{Sum of Squares}}{\text{degrees of freedom}}$

df between Groups $= k-1 = 3-1 = 2$ (3 groups)

df within Groups $= N-k = 15-3 = 12$ (15 sample size)

M. Squares b/w Groups $= 6.977/2 = 3.4885$

Mean Squares within Groups $= 0.499/12 = 0.0416$

* F statistic $= \dfrac{MS_B}{MS_W} = \dfrac{3.4885}{0.0416} = 83.86$

F value at $\left.\begin{array}{l}\alpha = 0.05 \\ df_1 = 2,\ df_2 = 12\end{array}\right\}$ 3.89

So our calculated F statistic > F critical ($\because$ 83 > 3.89)

So Reject H_0

$\therefore$ At 5% significance level, mean customer satisfaction scores are different b/w atleast 2 or more groups. (H_a)

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(Q4): House Price prediction data is given.

Calculate Pearson Correlation coefficient b/w 2 features given?

And interpret results?

$r = +1 \Rightarrow$ strongly positive correlation } Helps prediction

$r = -1 \Rightarrow$ strongly Negative correlation } Helps prediction

$r = 0$ (Near) $\Rightarrow$ No relation b/w features } doesn't help prediction

$$\boxed{r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}}$$

Pearson Correlation Coefficient

Given Data:

$\bar{x} = 3+4+2+5+3+4/6 = 21/6 = 3.5$

$\bar{y} = 250+320+200+400+260+310/6 = 1740/6 = 290$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - x)^2$	$(y_i - \bar{y})^2$	$(x_i - x)(y_i - \bar{y})$
3	250	-0.5	-40	0.25	1600	20
4	320	+0.5	+30	0.25	900	15
2	200	-1.5	-90	2.25	8100	135
5	400	+1.5	+110	2.25	12100	165
3	260	-0.5	-30	0.25	900	15
4	310	+0.5	+20	0.25	400	10
				5.5	24000	360

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$$r = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\Sigma(x_i-\bar{x}_i)^2 \, \Sigma(y_i-\bar{y})^2}} = \frac{360}{\sqrt{(5.5)(24000)}} = \frac{360}{\sqrt{132000}} = 0.990867$$

$r = 0.990867$

→ close to +1.

∴ No. of room increases Price increases. Relation is +99%.

∴ No. of rooms decrease Price decreases too.

It has very strong linear relationship.

That's why "No. of Rooms" is a very good predictor for House prices.

for given sample data.

End